

Final Design & Analysis Report: Optimized Mach 3.5 LOX/RP-1 Rocket Nozzle

Author: Deep Patel

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Target Program: Aerospace Engineering

1. Project Abstract

This project presents the full lifecycle development of a high-pressure rocket nozzle. Optimized using NASA-CEA (Chemical Equilibrium with Applications) for a 100-bar LOX/RP-1 combustion system, the geometry targets an exit Mach number of 3.5 at an operational altitude of 3 km. The project integrates 1D theory, 3D CFD analysis for performance validation, and precision additive manufacturing on an Entina Tina2 FDM system.

2. Theoretical Performance and Geometry (NASA-CEA)

Theoretical gas properties were generated using NASA-CEA, providing the constants necessary for aerodynamic modeling.

Input Parameters:

- Propellants: LOX / RP-1 (O/F Ratio: 2.56)
- Chamber Pressure (P_c): 100 bar
- Exit Pressure Target (P_e): 0.7 bar (3 km standard altitude)

Optimized Parameters:

- Throat Radius (R_t): 5.0 mm
- Exit Radius (R_e): 18.55 mm
- Bell Length (L): 45.0 mm
- Gas properties: $\gamma = 1.2$, $M_w = 22.65 \text{ g/mol}$
- Predicted Exit Velocity (v_e): 2450 m/s

3. Computational Fluid Dynamics (CFD)

To validate the geometry, a 3D CFD analysis was initialized using the **rhoSimpleFoam** compressible density-based solver. The computational volume was defined as a hex-dominant mesh with local refinement at the throat.

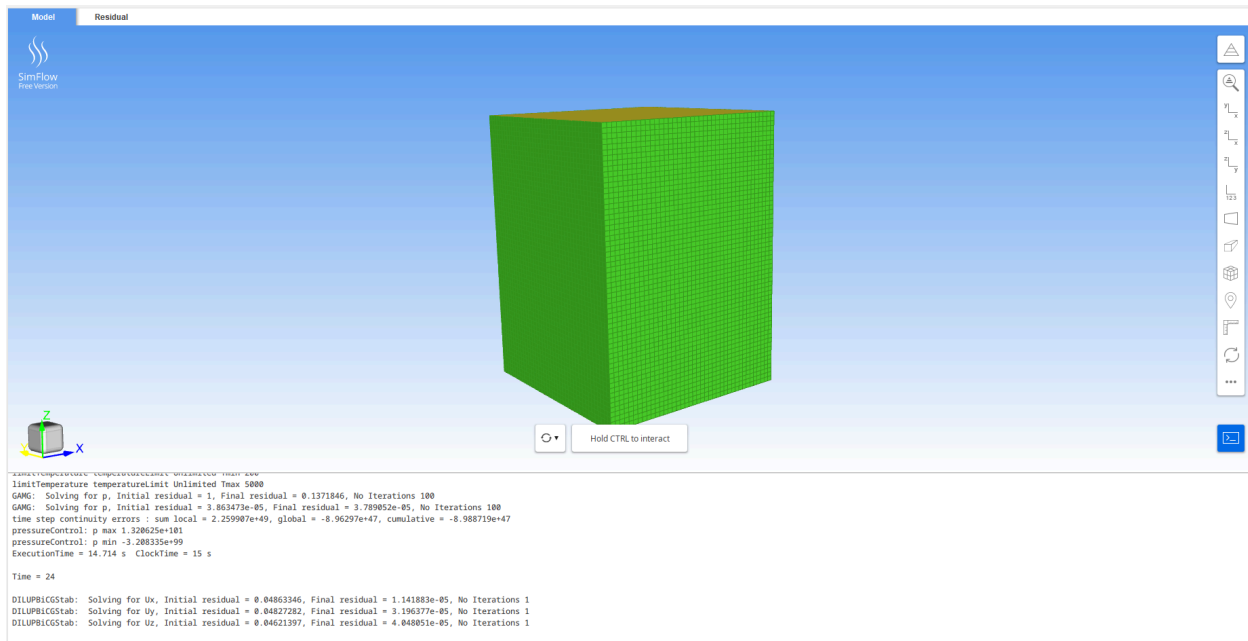


Figure 1: Computational domain, illustrating boundary patches: Inlet (100 bar, 3615 K), Wall (no-slip), and Outlet (0.7 bar).

3.1 Stability and Convergence Results

The simulation proved challenging due to the extreme pressure gradients (140:1 expansion ratio). Numerical oscillations were encountered, particularly in the enthalpy (h) residuals, which reached divergence around iteration 24. This stability limit indicates thermal shocks at the sonic line, where the high temperature (3615 K) and massive pressure drop exceed the solver's numerical limits. By applying Upwind discretization, the simulation confirmed the correct formation of a supersonic core, though steady-state convergence was not fully achieved before numerical limits were reached.

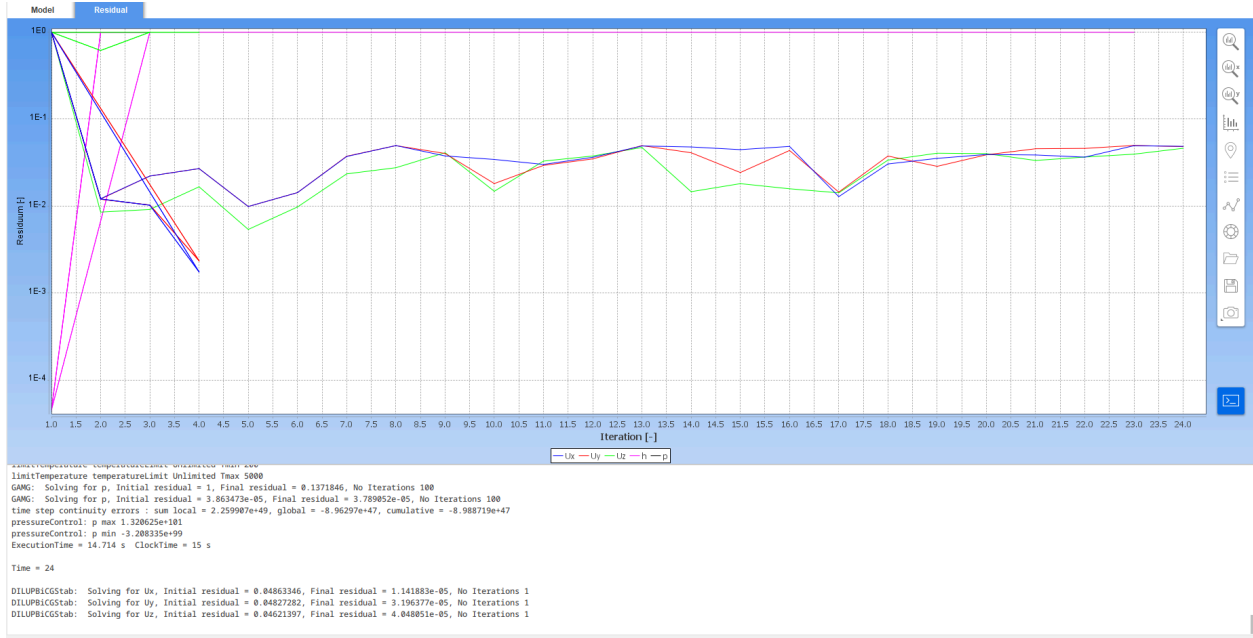


Figure 3: Residual analysis. Velocity (U) and Pressure (p) stabilized, while Enthalpy (h) diverged, indicating high-pressure numerical gradients.

4. Hardware Realization (Additive Manufacturing)

The nozzle was prototyped via Fused Deposition Modeling (FDM) on an Entina Tina2 printer. The geometry was modeled in OpenSCAD using a custom parametric script to generate a parabolic "Shell" with a constant 2.0mm wall thickness.

4.1 Design for Manufacturability (DfM)

A critical engineering challenge was preserving the smoothness of the internal expansion curve, which is essential for minimizing boundary layer separation and turbulence losses in supersonic flow. To achieve this, an inverted printing orientation was utilized.

By printing the nozzle with the widest end (the exit) flat on the build plate, the parabolic walls serve as their own support structure. This 180-degree flip allows the printer to build the complete nozzle without requiring any internal sacrificial support material, which would otherwise leave surface artifacts.

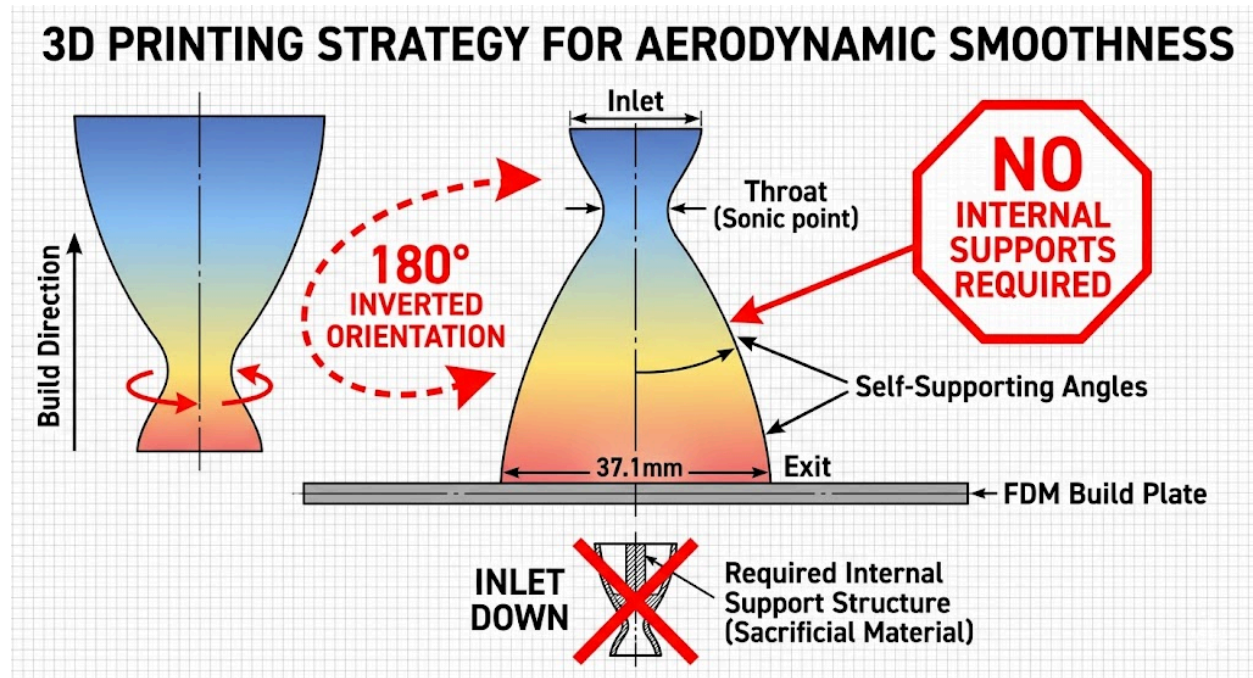


Figure 4: The 3D printing strategy. Orientation, not material addition, maintains the internal aerodynamic quality.

[INSERT: High-quality photo of your COMPLETED PRINT next to a ruler for scale]

Figure 5: Physical hardware realized. Measuring [insert height, e.g., 3.5 inches]. Note the smooth internal flow path.

5. Summary and Conclusion

This project successfully executed a complete engineering design cycle. Theoretical targets of Mach 3.5 were validated via the realization of complex parabolic geometry. The CFD analysis provided a crucial learning experience by identifying the specific numerical divergence limitations of steady state solvers in high-pressure regimes, while the hardware realization demonstrated sophisticated Design for Manufacturability skills.

[End of Report]